Given: Due:	Further Maths (Core Pure) HW 25	Name:

Chapter 9: Vector lines, planes & distance

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)

Useful resources: Your notes from recent lessons & your text book

Questions - do these on lined paper

Q1. The line *l* has equation

$$\frac{x+2}{1} = \frac{y-5}{-1} = \frac{z-4}{-3}$$

The plane *Π* has equation

$$\mathbf{r}.(\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = -7$$

Determine whether the line *l* intersects *Π* at a single point, or lies in *Π*, or is parallel to *Π* without intersecting it.

(Total for question = 5 marks)

Q2. The line l passes through the point *P*(2,1,3) and is perpendicular to the plane *Π* whose vector equation is

$$\mathbf{r}.(\mathbf{i} - 2\mathbf{j} - \mathbf{k}) = 3$$

Find

(a) a vector equation of the line *l*,

(2)

(b) the position vector of the point where *l* meets *Π*.

(4)

(c) Hence find the perpendicular distance of *P* from *Π*.

(2)

Q3. With respect to a fixed origin *O*, the line *l* has equation

$$\mathbf{r} = \begin{pmatrix} 13 \\ 8 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \text{ where } \lambda \text{ is a scalar parameter.}$$

The point *A* lies on *l* and has coordinates (3, − 2, 6).

The point *P* has position vector ($-p\mathbf{i} + 2p\mathbf{k}$) relative to *O*, where *p* is a constant.

Given that vector $\overrightarrow{PA}$ is perpendicular to *l*,

(a) find the value of *p*.

(4)

Given also that *B* is a point on *l* such that ∠*BPA* = 45°,

(b) find the coordinates of the two possible positions of *B*.

(5)

Q4. The plane Π_1 has vector equation $\mathbf{r}.\begin{pmatrix}2\\1\\3\end{pmatrix} = 5$

The plane Π_2 has vector equation $\mathbf{r}.\begin{pmatrix}-1\\2\\4\end{pmatrix} = 7$

(a) Find a vector equation for the line of intersection of Π_1 and Π_2, giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors and λ is a scalar parameter.

(6)

The plane Π_3 has cartesian equation

$$x - y + 2z = 31$$

(b) Using your answer to part (a), or otherwise, find the coordinates of the point of intersection of the planes Π_1, Π_2 and Π_3

(3)

Q5. Part of the mains water system for a housing estate consists of water pipes buried beneath the ground surface. The water pipes are modelled as straight line segments. One water pipe, W, is buried beneath a particular road. With respect to a fixed origin O, the road surface is modelled as a plane with equation $3x - 5y - 18z = 7$, and W passes through the points $A(-1, -1, -3)$ and $B(1, 2, -3)$. The units are in metres.

(a) Use the model to calculate the acute angle between W and the road surface.

(5)

A point $C(-1, -2, 0)$ lies on the road. A section of water pipe needs to be connected to W from C.

(b) Using the model, find, to the nearest cm, the shortest length of pipe needed to connect C to W.

(6)

END OF HOMEWORK
Total = 42 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.